

ACADEMIC DETAILS

Degree/Certificate	Specialization	Institute/Board	Year	CPI/%
B.Tech.	Mechanical Engineering	Indian Institute of Technology Gandhinagar	2024–2028	8.70
Senior Secondary	Physics, Chemistry, Mathematics	Late P. B. Jog Junior College (MSBSHSE)	2024	89.5
Secondary	SSC	Mahesh Vidyalaya English Medium School	2022	95.8

PROJECTS

- **Neural Network Framework for Learning Constitutive Laws in 2D Granular Flow**

[Jan 2026 – Apr 2026]

(Prof. Harkirat Singh, IIT Gandhinagar) | Project Link

- Engineered a modular PyTorch deep learning framework, evaluating 3 critical internal stress components (σ_{xx} , σ_{yy} , σ_{xy}) across 25 hyperparameter configurations to accurately map nonlinear physics paths.
 - Optimized overall cross-dataset prediction error rates from 48% down to 3% by implementing math-based reference normalization models, confirming robust generalization on 2 unseen datasets.
 - Streamlined the data preprocessing and training pipeline by automating 2D spatial grid mapping routines to handle large-scale internal 2D stress tensor variations.
- **Four-Bar Mechanism Kinematic Analysis**

[Aug 2025 – Nov 2025]

(Course Project, IIT Gandhinagar) | Project Link

- Modelled a planar 4-bar linkage using vector-loop kinematics to dynamically map and compute the displacement, velocity, and acceleration parameters of all moving links throughout a complete motion cycle.
 - Validated analytical vector loops against experimental profiles using high-speed 211.55 FPS video motion tracking to isolate deviations caused by joint clearances and measurement uncertainty.
 - Evaluated coupler-link rigid body kinetics across 12 distinct crank orientation geometries, resolving relative velocity profiles ($\vec{V}_G$) and angular velocities (Ω) from manual point-mass trajectory data.
- **Validation of Angular Momentum Conservation via Reynolds Transport Theorem**

[Jan 2026 – Apr 2026]

(Prof. Shubhangi Bansude, IIT Gandhinagar) | Project Link

- Validated physical angular momentum verification milestones by fabricating a custom 3-nozzle rotating water sprinkler mechanism prototype from scratch following formal mathematical validation rules.
 - Modelled relative error variances of 31.1% and 42.0% across 2 distinct operating speeds by formulating a structural fluid predictive framework balancing exit velocity and rotational speed metrics.

TECHNICAL SKILLS

- Programming Languages: C, C++, Python
- Libraries/Frameworks: NumPy, Pandas, Matplotlib, PyTorch (Basic)
- Developer Tools: Git, GitHub, Visual Studio Code
- Software & Simulation Utilities: MATLAB, Cantera, Mathematica, Autodesk Inventor, MSC Adams (Basic), ANSYS Fluent (Basic), Creality Print

RELEVANT COURSEWORK

- Mechanical Engineering:** Thermodynamics, Fluid Dynamics, Mechanics of Solids, Statics and Dynamics, Principles of Manufacturing Processes, Vibrations

ACHIEVEMENTS

- Awarded the Dean’s List (top 5% of class) and secured Rank 1 (Branch Topper) in Mechanical Engineering with an SPI of 9.48 in Semester 3.
- Awarded an A+ (11/10) grade in Thermodynamics for exceptional academic performance and accuracy across advanced curriculum evaluation structures.
- Contributed as a team member of the 8-member IIT Gandhinagar squad to secure First Position out of 250+ teams in Division 4 of the India College Chess Championship 2025, claiming a INR 60,000 prize.
- Secured Team Runner-up honors at the DA-IICT Chess Tournament and individual Runner-up (Unrated Category) recognition at the competitive IIT Bombay Sports Fest.

POSITIONS OF RESPONSIBILITY

- **Coordinating Team Member, Institute Sports Events, IIT Gandhinagar**

[Jul 2024 – Apr 2026]

- Managed basic operational logistics, on-ground scheduling configurations, and formal tracker loops for sports festivals while collaborating closely with student council representatives to facilitate tournament timelines.
- **Junior Team Member, Amalthea & Blithchron Festivals, IIT Gandhinagar**

[Aug 2024 – Feb 2025]

- Coordinated core event operations during Amalthea’25 by assisting with hospitality logistics and tracked company outreach alongside record matrices to maintain partner communications for Blithchron’25.

EXTRA-CURRICULAR ACTIVITIES

- Represented IIT Gandhinagar twice consecutively at the national Inter IIT Sports Meet, competing in Chess matches against varsity chapters from over 20 participating IITs.
- Participated actively in campus tournament brackets and regional inter-university chess invitations hosted at external venues including DA-IICT, Ahmedabad University, and IIT Jodhpur.